COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 6 : SLO 2 — Carry-Save Adders
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1. Understanding the Core Concept (Remember)

• Answer: b) To add three or more binary numbers by separating the sum and carry bits.

2. The 'Save' in Carry-Save (Understand)

• It is called carry-save addition because the carry bits are not propagated immediately to the next more significant bit; they are 'saved' and computed separately from the sum bits.

• Each stage takes three input bits and produces a sum bit and a carry-out bit, avoiding the delay of waiting for a carry to ripple through the adder.

3. Optimizing Multiplication (Apply)

• Step 1 — Generate the partial products of the 4-bit multiplication.

• Step 2 — First CSA stage: take three partial products and add them with a layer of carry-save adders, producing a sum vector and a carry vector.

• Step 3 — Subsequent CSA stages: feed the previous sum and carry vectors plus the next partial product into another CSA layer; repeat until only two vectors (one sum, one carry) remain.

• Step 4 — Final addition: add the final sum and carry vectors with a conventional carry-propagate adder (ripple-carry or carry-lookahead) to obtain the product.

4. Analyzing the Advantage (Analyze)

• With traditional ripple-carry adders, every addition has a delay proportional to the number of bits because the carry ripples from LSB to MSB, and these delays accumulate over multiple additions.

• A carry-save adder tree adds three numbers in constant time regardless of bit-width, because carries are not propagated; the single carry-propagate addition happens only once at the very end.

• Key advantage: a significant speed improvement — the expensive carry propagation is postponed to one final step, making carry-save addition much faster for multi-operand scenarios such as multiplication.